Chun Song

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Education

The Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, P.R. China*
FULLY FUNDED PH.D. STUDENT, URBAN GOVERNANCE AND DESIGN (UGOD) — *Sep. 2025 - Present*

Institute of Engineering Mechanics, China Earthquake Administration — *Harbin, P.R. China*
MASTER OF ENGINEERING IN STRUCTURAL ENGINEERING — *2022 - 2025*

Beijing Jiaotong University — *Beijing, P.R. China*
B.E. IN CIVIL ENGINEERING, SCHOOL OF CIVIL ARCHITECTURE AND ENGINEERING — *2018 - 2022*

Research Interests

`Core Research Interests:` Multi-agent systems, disaster simulation, and crowd dynamics.

`Core Research Question:` How can we develop generalizable, verifiable simulations and effectively integrate AI to enable prediction, optimization, and decision-making?

Research Affiliations

Responsible Urban Intelligence Lab (RUI Lab), HKUST(GZ) — *Guangzhou, P.R. China*
RESEARCH ASSISTANT, SUPERVISED BY PROF. RUI CAO — *Jul. 2025 - Aug. 2025*

Chongqing University Liyang Smart City Research Institute — *Liyang, P.R. China*
RESEARCH STUDY — *2023 - 2025*

Harbin Institute of Technology, School of Civil Engineering — *Harbin, P.R. China*
RESEARCH STUDY — *2022 - 2023*

Research Funding

Guangdong Provincial Graduate Student Research Project — *2026*

- **Project Lead**; RMB 20,000 awarded for "Physics-Simulation-Driven Digital-Twin Scenario Deduction and Emergency Decision-Making Methods for Community Indoor Fires."

Honors and Awards (Selected)

2026 — **Third Prize (two projects)**, Urban Cup 2026. Award List
- "Governing the AI Shock: From Laissez-Faire Disruption to a Broad-Based AI Social Compact." Project Poster | Competition Video
- "Personal Shade, Shared Consequences: How Umbrellas Reshape Movement and Solar Exposure in Dense Public Spaces." Project Poster.

2025 — **Excellent Simulation Technology Solution**, 2025 Typical Case Collection, for "Large-scale Multi-hazard Digital Simulation and Scenario-deduction System for Cities and Urban Agglomerations." Certificate | Project Poster.

2025 — **Excellent Presentation Award**, the 7th National Conference on Disaster Prevention and Mitigation Engineering, for "Urban Emergency Evacuation Simulation Models and Systems." Conference Program | Certificate.

2024 — **Second Prize**, Postgraduate Division, the 4th Engineering Simulation Innovation Design Competition, 2024 Belt and Road and BRICS Skills Development and Technology Innovation Competition. Certificate.

2023, 2024 — **First-class Graduate Academic Scholarship**, Institute of Engineering Mechanics (two-time recipient).

2022 — **Second-class Graduate Academic Scholarship**, Institute of Engineering Mechanics.

2021 — **Honorable Mention**, 2021 Interdisciplinary Contest in Modeling (ICM). Team: Yingying Han, Yuqin Zhou, and Chun Song; faculty advisor: Guozhong Liu. Certificate.

Publications († corresponding author)

[P5] A Simulation-Based Framework for Optimizing Road Networks for Disaster Evacuation in Dense Informal Urban Settlements.
Chun Song, Zhijie Zhou, Rui Cao†.
Under review at Computers, Environment and Urban Systems (CEUS). Poster.

[P4] Unveiling Inequalities in Forgotten Enclaves: A Geospatial-Streetscape Integrated Framework for Assessing Urban Village Livability.
Zhijie Zhou, Chun Song, Rui Cao†.
Under review at Cities.

[P3] Coupled Density-Hazard Navigation for Evacuation Reliability in Dense Urban Informal Settlements.
Chun Song, Xuchuan Lin, Rui Cao†.
Reliability Engineering & System Safety, 2026. Paper | Poster.

[P2] Analysis of influencing parameters of building fire spread simulation in urban area based on cellular automata.
Ning Yang, Qi Wang, Jiangrong Zhong, Xuchuan Lin†, Chun Song.
The 18th China CAE Engineering Analysis Technology Annual Meeting and 14th China Digital Simulation Forum, in Chinese.

[P1] Limited Monitoring Data Based Updating Method for Fundamental Period of the Building Group Model.
Chen Xianan, Lin Xuchuan†, Zhang Lingxin, Gao Wuping, Song Chun.
Engineering Mechanics, 2024, in Chinese.

Conference Presentations († corresponding author)

[Poster 2] Coupled Density-Hazard Navigation for Evacuation Reliability in Dense Urban Informal Settlements.
Chun Song, Xuchuan Lin, Rui Cao†.
Poster accepted at the 4th International Conference on Urban Science and Intelligence, Guangzhou, China, August 8-9, 2026. Conference | Poster.

[Poster 1] A Simulation-Based Framework for Optimizing Road Networks for Disaster Evacuation in Dense Informal Urban Settlements.
Chun Song, Zhijie Zhou, Rui Cao†.
Poster accepted at the 6th CAAI Symposium on Social Computing and Social Intelligence (SCIISummit 2026), Guangzhou, China, April 17-18, 2026. Conference Report | Poster.

[Oral 1] Modeling and Simulation of Urban Crowd Emergency Evacuation.
Chun Song, Yue Wang, Xuchuan Lin†.
Oral presentation at the 3rd Academic Conference on Digital Twin and Intelligent Construction, December 15, 2024. Presentation Certificate.

Intellectual Property

[Patent 2] Road-widening optimization method, system, and storage medium driven by emergency-evacuation simulation diagnosis.
Rui Cao, Chun Song, Zhijie Zhou.
Chinese invention patent, granted June 9, 2026. Patent No. ZL 2026 1 0338721.6. Patent Certificate.

[Patent 1] Simulation method, system and computer storage medium of earthquake secondary fire spread.
Xuchuan Lin, Qi Wang, Jiangrong Zhong, Chun Song, Ning Yang.
Published.

[Software Copyright 1] Crowd Emergency Evacuation Simulation Software V1.0.
Chun Song, Xuchuan Lin, Jiangrong Zhong, Qi Wang, Xianan Chen, Xiaoxiang Yuan, Xiaonan Zhang.
Computer software copyright, registered November 19, 2024. Registration No. 2024SR1825682. Certificate.

Skills

Programming	Python, C++.
Research	Physics-based simulation modeling, GIS, reinforcement learning.

Academic Services

2026-27 Fall — Graduate Teaching Assistant, *Introduction to GeoAI* (UGOD 5300), The Hong Kong University of Science and Technology (Guangzhou).

2025-26 Spring — Teaching Assistant for Undergraduate Course, *History of Chinese Civilization: From Antiquity to Late Imperial* (UCUG 1604), The Hong Kong University of Science and Technology (Guangzhou).